Rajalaxmi Rajagopalan

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Education

University of Illinois, Urbana-Champaign **Fall 2021 - Spring 2026**
PhD, ECE. *GPA 4.0*
Relevant Coursework - Optimization, vector space signal processing/optimization, Random processes, Machine Learning for Signal Processing, Information Theory, Statistical Inference

National university of Singapore **Fall 2018 - Spring 2020**
Master of Science *GPA 4.45/5*
Department of Electrical & Computer Engineering

CEG, Anna University, India **Fall 2012 - Spring 2016**
Bachelors in Engineering *GPA 9.75/10*
Department of Electrical & Computer Engineering

Publications

[1] **R. Rajagopalan**, Debottam Dutta, Y.L. Wei, and R.R. Choudhury, "Personalized Image Generation via Human-in-the-loop Bayesian Optimization", **ICML**, July 2026.

[2] Debottam Dutta, Jianchong Chen, **R. Rajagopalan**, Y.L. Wei, and R.R. Choudhury, "CO3: Contrasting Concepts Compose Better in Diffusion Models", **ICLR**, April 2026.

[3] **R. Rajagopalan**, R. Giri, Z. Tang, K. Han, and G. Friedland, "Masked Autoencoders as Universal Speech Enhancer", **ICASSP**, April 2025.

[4] **R. Rajagopalan**, Y.L. Wei, and R.R. Choudhury, "Kernel Learning for Sample Constrained Black-Box Optimization", **AAAI**, February 2025.

[5] **R. Rajagopalan**, Y.L. Wei, and R.R. Choudhury, "Sample-Constrained Black Box Optimization for Audio Personalization", **AAAI**, February 2024.

[6] **R. Rajagopalan**, Y.L. Wei, and R.R. Choudhury, "Audio Personalization through Human-in-the-loop Optimization", **NeurIPS Workshop on ML for Audio**, December 2023.

[7] Y.L. Wei, **R. Rajagopalan**, B. Islam, and R.R. Choudhury, "Self-supervised Speech Enhancement using Multi-Modal Data", **NeurIPS Workshop on ML for Audio**, December 2023.

[8] **R. Rajagopalan** and C. K. Tham, "Active Learning for IoT Data Prioritization in Edge Nodes Over Wireless Networks," IECON 2020 The 46th Annual Conference of the IEEE Industrial Electronics Society, Singapore, 2020.

Honors and Awards

■ **A.R. "Buck" Knight Fellowship**	UIUC	Fall 2025
■ **Amazon-Illinois (AICE) Fellowship**	UIUC	Spring 2025
■ **Exceptional Academics Award**	National University of Singapore	Spring 2020
■ **Silver Medal- Department second rank award**	CEG, Anna University	Spring 2016
■ **Gold Medal- Excellence in Mathematics**	CEG, Anna University	Spring 2013

Research Experience

Graduate Student Researcher Fall 2021 - Present
Signals & Inference Research Group (SiNRG) `https://sinrg.csl.illinois.edu/` UIUC
- Sample Constrained Black-Box Optimization using techniques like **Bayesian Optimization Gaussian Process Regression** for Human-in-the-loop personalization of content: audio, images, etc.
- Multi-modal universal speech enhancement through self-supervised learning.
- Other projects in acoustic signal processing and earable computing.

Graduate Student Researcher Fall 2018 - Spring 2021
Wireless Networking Group NUS, Singapore
- **Distributed modeling** and **Bayesian statistical inference** in resource-constrained large-scale distributed parallel computing areas: wireless sensor networks and IoT.

Undergraduate Student Researcher Spring 2014 - Spring 2016
Integrated Systems Lab (ISL) CEG, India
- ANUSAT-2 satellite (QB50 European Research Mission) https://www.qb50.eu/index.php/news.html

Work Experience

Applied Scientist Present
AWS AI Labs

Applied Scientist Intern Summer 2024, 2025
AWS AI Labs
- Speech and audio-related problems in the Transcribe AWS team.
- Self-Supervised Universal Speech Enhancement.
- Music Generation via Inference-time Optimization of Diffusion Models.

Teaching Assistant Spring 2023
UIUC

Research Assistant Fall 2021 - Spring 2026
SiNRG, UIUC
- Conducting research in audio signal processing and black-box optimization.

Teaching Instructor Spring 2020 - Spring 2021
NUS Singapore

Network Planner - Assistant manager June 2016 - July 2018
Airtel Gurgaon, India

Skills

- **Languages :** Python, R, C, C++, VHDL, MATLAB, VBA
- **Hardware Platforms :** USRP, Raspberry PI, BlackFin BF609, BladeRF, SDR, Cyclone –II FPGA, Stratix-III FPGA, TIVA.
- **Certifications:** Cisco Network Master
- **Languages:** English, Tamil, Hindi, French, Japanese
- **Extracurriculars:** Trained Indian classical violinist

Selected Projects

Sample Constrained Black-Box Optimization for Personalization *2021-Present*
Prof Dr. Romit Roy Choudhury - SiNRG, UIUC
- We consider the problem of personalizing content to a user's taste. Content could be audio signals in a hearing aid, customized images, etc. Given a content generator space, our goal is to find the optimal sample that will maximize the user's personal satisfaction.The user satisfaction function is unknown, and hence optimizing it using the human in the loop, while constrained by the number of samples, is a sample-constrained black-box optimization problem. We build on the Bayesian Optimization framework. We achieve sample efficiency by developing kernel learning techniques that learn a unique kernel that models the function structure.

MusicX - Guided Music Generation via Diffusion Inference-time Optimization *Summer 2025*
Turab Iqbal, Erfan Soltanmohammadi - AWS AI Labs
- We propose MusicX, a training-free, fast, and versatile music generator that modifies music generated by a generic frozen text-to-audio latent diffusion model to generate new music with specific characteristics that the frozen model is incapable of by, (1) Guidance Control: that modifies the diffusion trajectories through an optimization in the diffusion latent space, enabling training-free modification for arbitrary conditioning modalities. (2) Fused Diffusion: merges multiple guided diffusion trajectories, enabling variable-length multi-conditioned generation. (3) Fast Distill: speeds up the distillation by 1000x by modifying the CTM distillation scheme.

Self-supervised Universal Speech Enhancement *Summer 2024*
Ritwik Giri, Zhiqiang Tang - AWS AI Labs
- Speech is often corrupted by many distortions like ambient noise, reverberation, limited bandwidth, codec artifacts, interfering speakers etc. In practical scenarios, there is lack of high-quality clean speech references. Thus, we developed a universal speech enhancer that (1) can tackle a broad range of distortions simultaneously, (2) trained in a self-supervised manner without access to large-scale clean speech corpus, and (3) offers robust performance for enhancement downstream tasks.

Self Supervised IMU-based Speech Enhancement *2021-Present*
Prof Dr. Romit Roy Choudhury - SiNRG, UIUC
- Extracting compact representations of personalized speech features like base frequencies using representation models like autoencoders from captured surface vibrations over the face from the speakers' throat by IMUs in earphones, to perform personalized denoising of speech corrupted by over-the-air interference.

Master's Thesis: Active Learning for IoT Data Prioritization in Edge nodes over Wireless Networks *2019*
Prof Dr. Chen-Khong Tham - WICOMM, NUS
- Developed Active learning-based and Bayesian ML-based techniques to improve the robustness of distributed deep learning models in the presence of practical wireless communication channel issues for the application of predictive maintenance of industrial machines.

Design of Anusat-2 Nano-Satellite (QB50 European Research Mission) *2015*
Prof Dr. P. V. Ramakrishna - ISL, CEG
- Designed and developed communication modules in a student-led CubeSat satellite.